

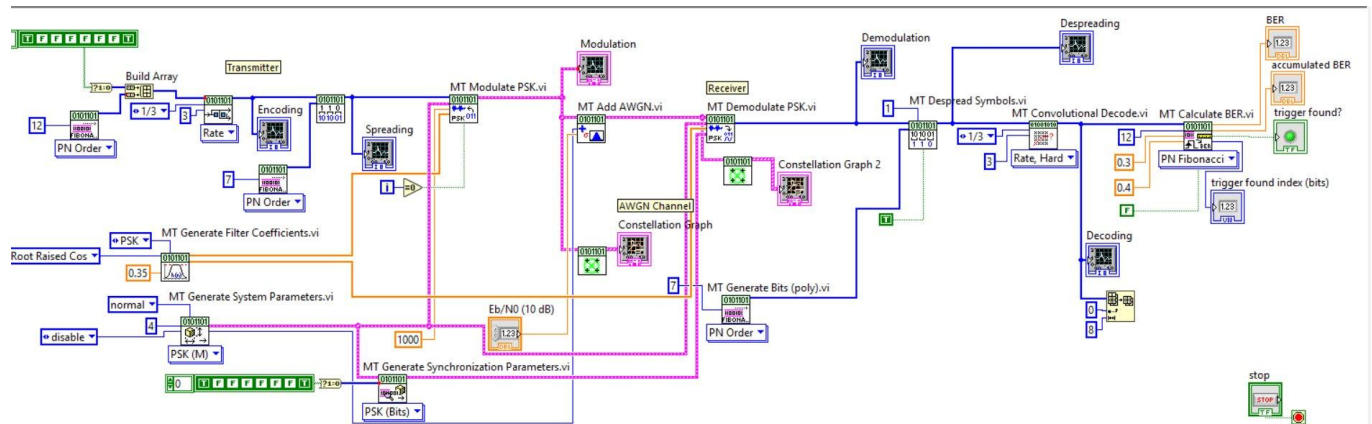
WCDMA System Implementation

Using LabVIEW and USRP Hardware

Baseband Processing, Spreading, Modulation & BER Analysis

SYSTEM BLOCK DIAGRAM

Figure 1: WCDMA System LabVIEW Block Diagram



1. INTRODUCTION

WCDMA (Wideband Code Division Multiple Access) is a 3G wireless communication standard based on CDMA technology. It allows multiple users to transmit simultaneously over the same frequency band using unique spreading codes, enabling high data rates and efficient spectrum utilization.

Key system parameters:

- Bandwidth: 5 MHz
- Modulation: QPSK (Quadrature Phase Shift Keying)
- Channel Coding: Convolutional coding (rate 1/3, K = 3)
- Multiple Access: Code Division — each user assigned a unique PN spreading code

Spreading techniques increase resistance to interference and multipath fading. Despreading at the receiver, using the same PN code, recovers the original data. This project implements a simplified WCDMA system in **LabVIEW** using **USRP hardware**, covering all baseband processing stages and evaluating performance via **Bit Error Rate (BER)** analysis.

2. SYSTEM ARCHITECTURE

2.1 Block Diagram Overview

The WCDMA system consists of the following processing chain:

1. **Input Data Source** — Provides the original digital bit stream to transmit.
2. **Convolutional Encoder** — Adds redundancy (rate 1/3, K = 3) for error correction.

3. **Spreader** — Multiplies encoded bits with a PN sequence, widening the signal bandwidth.
4. **QPSK Modulator** — Maps bits onto four phase states; transmits 2 bits per symbol.
5. **USRP RF Transmission** — Upconverts and transmits the modulated signal over the air.
6. **QPSK Demodulator** — Converts the received RF signal back to baseband bits.
7. **Despreader** — Correlates the received signal with the same PN code to recover encoded data.
8. **Viterbi Decoder** — Applies the Viterbi algorithm to correct channel errors and recover original bits.

2.2 Multiple Access in CDMA

CDMA enables multiple users to share the same frequency band simultaneously:

- Each user's data is spread using a unique PN spreading code.
- Spreading codes are designed to be orthogonal or have very low cross-correlation to minimize inter-user interference.
- All users' signals are transmitted simultaneously over the same channel.
- At the receiver, the intended signal is recovered by correlating with the corresponding PN code.
- Signals from other users appear as low-level noise due to code orthogonality.
- This allows efficient spectrum use without dedicated frequency or time slots per user.

2.3 Orthogonal Spreading Sequences

Orthogonal spreading sequences guarantee **zero cross-correlation** between different users' codes, ensuring minimal inter-user interference.

- Walsh-Hadamard codes are the standard orthogonal codes used in WCDMA systems.
- Generated from Hadamard matrices — square matrices whose rows are mutually orthogonal.
- Each user is assigned a unique Walsh code (row) from the matrix.
- During spreading, user data is multiplied by its assigned Walsh code.
- At the receiver, despreading with the same Walsh code recovers the user's data without inter-user interference.

3. TRANSMITTER

3.1 Data Source — PRBS Generation (MT Generate Bits)

The data source is implemented using the **MT Generate Bits (PN)** block in LabVIEW, which generates a **Pseudo-Random Binary Sequence (PRBS)** using a **Linear Feedback Shift Register (LFSR)**.

- A specified LFSR order and initial state produce a deterministic, seemingly random bit stream of 0s and 1s.
- This simulates real digital data in a controlled, repeatable manner — essential for BER evaluation.
- The generated PN bits serve as the input message data for encoding, spreading, and modulation stages.

3.2 Channel Coding — Convolutional Encoder (MT Convolutional Code)

Channel coding is applied using the **MT Convolutional Code** block in LabVIEW, configured with:

- Constraint length $K = 3$ (3 memory elements in the shift register).
- Code rate = $1/3$ (3 encoded output bits per 1 input bit).
- This introduces controlled redundancy, significantly improving resilience to noise and interference — critical for reliable WCDMA transmission.

3.3 Spreading — MT Spread Symbols

The encoded bits are spread using the **MT Spread Symbols** block with a PN sequence of **order 5** (sequence length = $2^5 - 1 = 31$ chips).

- The spreading process widens signal bandwidth, improving interference resistance and enabling multi-user CDMA access.
- A higher spreading factor increases processing gain but reduces effective data rate.

PN Sequence Generator — Fibonacci LFSR

The PN spreading sequence is generated by a **Fibonacci LFSR**. Specific tap positions (e.g., [16, 14, 13, 11]) are XOR-ed to produce a feedback bit, which is shifted into the register each clock cycle, generating a pseudo-random output bit.

The feedback taps correspond to a primitive polynomial, ensuring the LFSR cycles through all $2^n - 1$ non-zero states — producing a maximal-length sequence (m-sequence).

Key autocorrelation property of m-sequences:

- Peak autocorrelation at zero lag ($\tau = 0$): $R(0) = N$ (sequence length).
- Low constant value of $-1/N$ at all other shifts ($\tau \neq 0$).
- This near-ideal autocorrelation enables accurate synchronization and minimizes inter-user interference in spread-spectrum systems.

3.4 QPSK Modulation — MT Modulate PSK

QPSK (Quadrature Phase Shift Keying) is used for digital modulation. It maps **2 bits per symbol** onto four phase states: 0° , 90° , 180° , 270° .

Why QPSK is used in WCDMA:

- Doubles the data rate compared to BPSK without increasing bandwidth.
- Good balance between spectral efficiency and implementation complexity.
- Standard modulation in WCDMA, LTE, and Wi-Fi.

LabVIEW implementation parameters (MT Modulate PSK):

- $M = 4$ (QPSK constellation order).
- Filter type: Root Raised Cosine (RRC).
- Filter parameters (symbol rate, samples per symbol, filter gain) optimized for USRP transmission.

RRC Pulse Shaping & Matched Filter

A key design feature in this system is that a **single MT Generate Filter Coefficients.vi** block (configured with Root Raised Cosine, $\alpha = 0.35$) supplies the **same filter coefficients to both the Tx modulator and the Rx demodulator**, forming a proper **matched filter pair**.

- At the Tx: the RRC filter shapes each QPSK symbol impulse into a smooth pulse, confining bandwidth to $(1 + \alpha) / 2T$ Hz and avoiding adjacent-channel interference.
- At the Rx: the identical RRC filter acts as the matched filter. Their cascade (RRC * RRC) produces a full Raised Cosine (RC) response — satisfying the Nyquist zero-ISI criterion at every sampling instant.
- The matched filter maximizes SNR at the decision point ($\text{SNR} = 2E_s / N_0$) — no other linear filter performs better in AWGN.
- Sharing the same coefficient block guarantees mathematical consistency: any mismatch in α or filter length would break the matched condition, degrading SNR and introducing residual ISI.
- $\alpha = 0.35$: the WCDMA standard trade-off — higher α gives better ISI immunity; lower α gives tighter bandwidth but demands more precise timing.

3.5 RF Transmission via USRP

After baseband processing, the modulated signal is upconverted and transmitted over the air via the USRP hardware. The USRP performs digital-to-analog conversion, frequency upconversion, and RF amplification before transmission.

4. RECEIVER

4.1 QPSK Demodulation — MT Demodulate PSK with Matched Filtering

At the receiver, the **MT Demodulate PSK** block ($M = 4$) performs QPSK demodulation. Critically, it receives the **same RRC filter coefficients from MT Generate Filter Coefficients.vi** that were used at the transmitter — completing the matched filter pair described in Section 3.4.

- The block applies the RRC matched filter to the incoming signal before making symbol decisions.
- This maximizes SNR at the decision point ($2E_s / N_0$) and eliminates ISI via the combined RC response.
- Symbol decisions are made by comparing the filtered signal to the four QPSK decision boundaries (at 45° , 135° , 225° , 315°).
- All parameters (symbol rate, samples-per-symbol, filter coefficients) must be identical to the Tx — guaranteed in this design by sharing the same coefficient source block.

4.2 Frame Synchronization via Preamble

Accurate timing synchronization is essential for correct symbol interpretation. A known **preamble sequence** (*10000001*) is prepended to each transmitted frame.

- The receiver searches for this unique preamble pattern in the incoming bit stream.
- On detection, the receiver aligns its symbol clock (start-of-frame synchronization).
- This establishes reliable demodulation and decoding of all subsequent data.

4.3 Despreading — MT Despread Symbols

After demodulation, the **MT Despread Symbols** block correlates the received signal with the same PN code used at the transmitter to recover the encoded data.

Correct despreading requires PN code synchronization between Tx and Rx, handled by two parameters:

- Sync Confidence Threshold — correlation-based threshold; when exceeded, synchronization is declared.
- Sync Search Depth = 31 bits — matches the non-repeating length of the order-5 PN sequence ($2^5 - 1 = 31$), ensuring synchronization is based on the full, unique cycle of the sequence.

Sliding Correlation for PN Synchronization

- The receiver slides its locally-generated PN code over the incoming signal one chip at a time.
- At each offset, the cross-correlation between the local PN code and incoming signal is computed.
- Misalignment → low correlation → incorrect despreading.
- Correct alignment → peak correlation → synchronization declared → despreading proceeds.
- Once the correlation exceeds the Sync Confidence Threshold, the PN generators at Tx and Rx are aligned.

4.4 Viterbi Decoding — MT Convolutional Decoder

The final stage applies the **Viterbi algorithm** (MT Convolutional Decoder) to decode the convolutionally encoded data:

- Uses trellis-based maximum likelihood sequence estimation (MLSE).
- Decoder parameters: constraint length $K = 3$, rate = $1/3$ — must match the encoder exactly.
- Corrects bit errors introduced by the channel and recovers the original transmitted bit stream.

5. PERFORMANCE RESULTS — BER VS. SNR

System performance is evaluated using **Bit Error Rate (BER)** as a function of **Signal-to-Noise Ratio (SNR)**. BER measures the fraction of incorrectly received bits; SNR quantifies signal strength relative to noise.

5.1 Effect of Low SNR

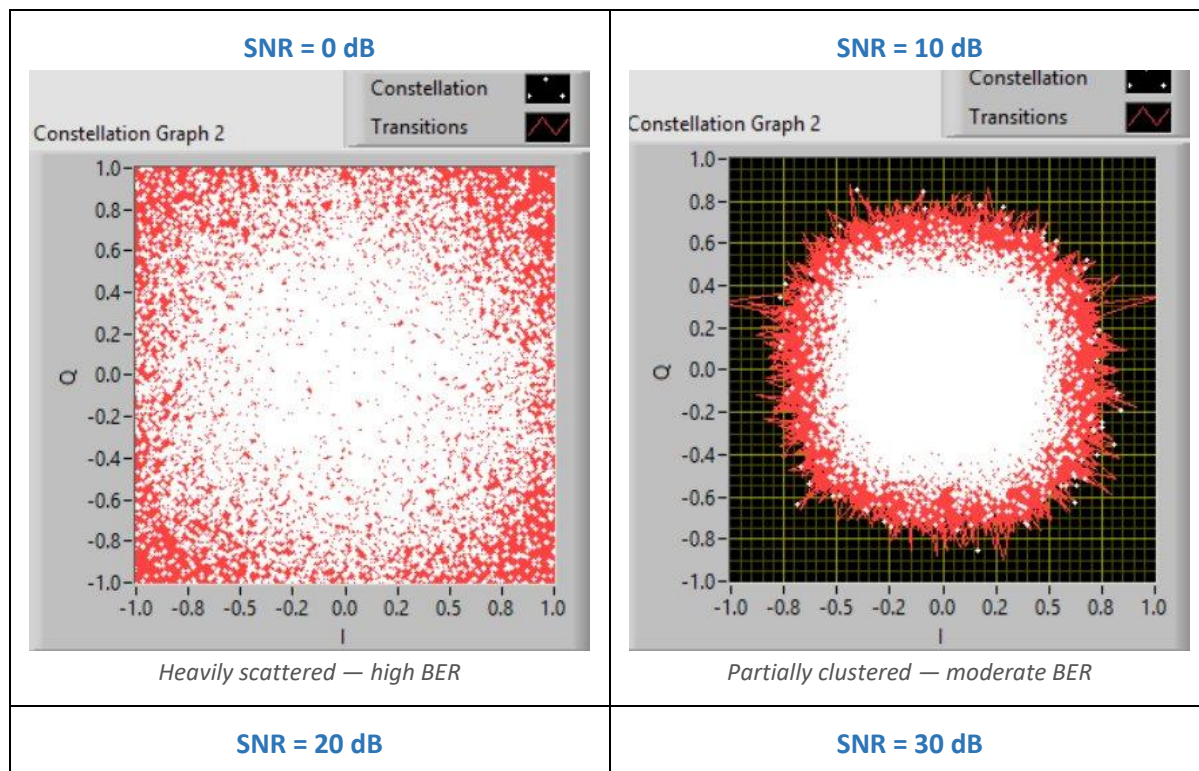
- Increased noise corrupts the received signal.
- QPSK constellation points become scattered — symbols drift closer to incorrect decision regions.
- Higher probability of symbol misclassification → BER increases.
- At very low SNR, nearly all output bits are in error.

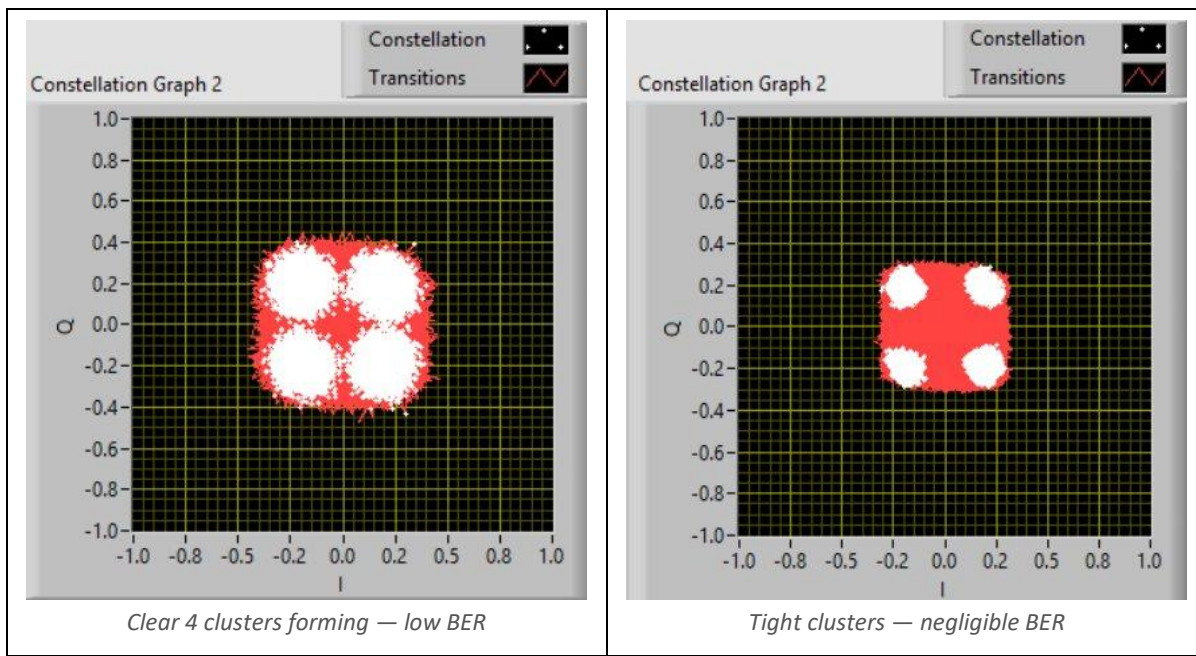
5.2 Effect of High SNR

- Reduced noise allows clearer signal recovery.
- QPSK constellation points cluster tightly around correct symbol positions.
- Fewer decoding errors → BER decreases significantly.
- At sufficiently high SNR, the Viterbi decoder recovers the original bit stream without any errors.

5.3 Constellation Diagrams

The following QPSK constellation diagrams show the received signal quality at four different E_b/N_0 (SNR) values. As SNR increases, the four QPSK symbol clusters become progressively more distinct, reflecting the reduction in bit errors.





The progression from SNR = 0 dB to SNR = 30 dB clearly demonstrates: **higher SNR produces tighter, more distinct QPSK symbol clusters, directly translating to lower BER and improved system reliability.**

5.4 Summary

Higher SNR → tighter constellation → lower BER → better system performance.

Lower SNR → scattered constellation → higher BER → degraded performance.

This BER vs. SNR relationship is the fundamental performance metric for digital communication systems and validates the correct operation of the WCDMA implementation.